



الجامعة السعودية الإلكترونية
SAUDI ELECTRONIC UNIVERSITY
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College of Computing and Informatics

Professional Ethics in IT



Module 9

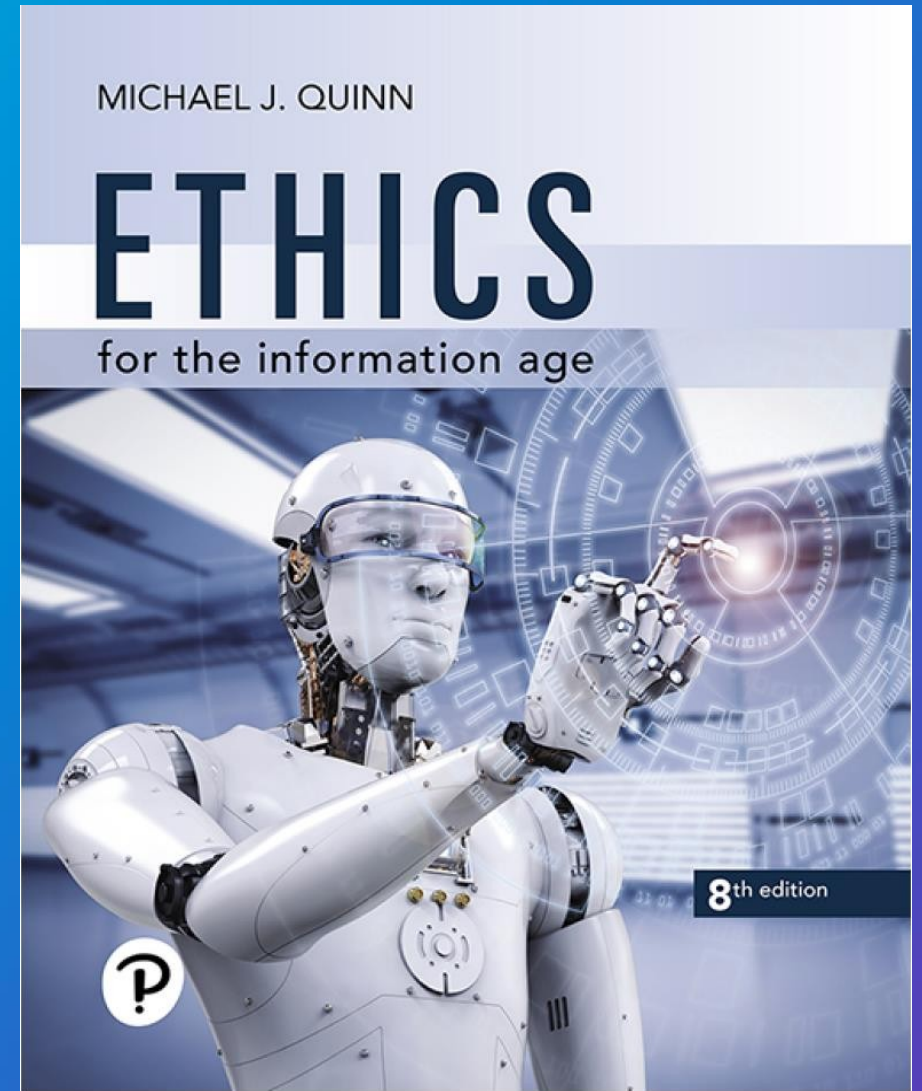
Chapter 7 - Computer and Network Security

by Quinn, M.J., 8th Edition



Required Reading

Chapter 7



Contents

- ❖ **Introduction**
- ❖ **Hacking**
- ❖ **Malware**
- ❖ **Cyber Crime and Cyber Attacks**
- ❖ **Online voting**



Learning Outcomes

- Identify techniques and practices used by hackers to gain unauthorized access to computers and computer networks.
- Recognize some of the illegal hacker-related activities.
- Identify the potential risks created by malware and the defensive measures that can protect computers from them.
- Evaluate ethical concerns related to computer and network security.
- Define common types of Cyber-Attacks.
- Identify techniques used by hackers to gain unauthorized access to computers and computer networks.



7.1 Introduction



Introduction

- Increased threats to computers and network security
- Increasing use of mobile computing
- The wide spread of open wireless network and potential issues
- Utility of networked computers increasing
 - Shopping and banking
 - Managing personal information
 - Controlling industrial processes
- The growing importance of computer security

Based on Quinn, M.J. (2020). *Ethics for the information age*. (8th ed.)

7.2 Hacking



7.2.1 Hackers, Past and Present

- Original meaning of hacker: explorer, risk taker, system innovator
 - Abounded MIT's Tech Model Railroad Club in 1950s and 1960
 - Labeling someone as a hacker was a sign of respect
- 1960s-1980s: Focus shifted from electronics to computers and networks
 - 1983 movie *WarGames*
- Modern meaning of hacker: someone who gains unauthorized access to computers and computer networks



7.2.2 Penalties for Hacking

- Hacker-related activities include – (Saudi Arabia Anti- Cyber Crime Law)
 - The law covers cybercrimes with penalties based on the consequence of them
 - Imprisonment penalties from one year to up to ten years
 - Fines up to SAR 5, 000,000
 - Attacks varies from unauthorized access with the intention of harm to unauthorized access with the intention to harm the national economy



7.2.3 Selected Hacking Incidents

- In 2003 a hacker broke into computers at the University of Kansas and copied the personal files of 1,450 foreign students
- MIT's Tech Model Railroad Club in 1950s
- 1960s-1980s: Focus shifted from electronics to computers and networks
 - 1983 movie *WarGames*
- Modern meaning of hacker: someone who gains unauthorized access to computers and computer networks



7.2.4 FBI and the Locked iPhone

- December 2015, Syed Rizwan Farook and Tashfeen Malik brought assault rifles and semiautomatic handguns into the Inland Regional Center in San Bernardino
- FBI recovered Farook's work-issued iPhone 5C and wanted to gain access of evidences
- FBI obtained a warrant but was not able to access the phone due the passcode
- The version of the iPhone iOS will encrypt the data if false attempts to access the are made
- In 2016, the FBI asked the cooperate in developing an iOS that disable the discussed restrictions

7.2.5 Case Study: Firesheep

- ❖ October 2010: Eric Butler released Firesheep extension to Firefox browser
- ❖ Firesheep made it possible for ordinary computer users to easily sidejack Web sessions
- ❖ More than 500,000 downloads in first week
- ❖ Attracted great deal of media attention
- ❖ Early 2011: Facebook and Twitter announced options to use their sites securely



Act-Utilitarian

Analysis

- ❖ Release of Firesheep led media to focus on security problem
- ❖ Benefits were high: a few months later Facebook and Twitter made their sites more secure
- ❖ Harms were minimal: no evidence that release of Firesheep caused big increase in identity theft or malicious pranks
- ❖ Conclusion: Release of Firesheep was good



Virtue-Ethics

❖ Analysis

- ❖ The creation of Firesheep was to make people aware of the problem that was ignored for too long
- ❖ All of the statements and actions showing that protecting the privacy of website visitors were that main intentions
- ❖ The tool was freely available online
- ❖ Therefore, from the perspective of *virtue ethics*, Butler's actions and statements were characteristic of someone interested in promoting the common good

Kantian

Analysis

- ❖ Accessing someone else's user account is an invasion of their privacy and is wrong
- ❖ Butler provided a tool that made it much simpler for people to do something that is wrong, so he has some moral accountability for their misdeeds
- ❖ Butler was willing to tolerate short-term increase in privacy violations in hope that media pressure would force Web retailers to add security
- ❖ He treated victims of Firesheep as a means to

7.3 Malware

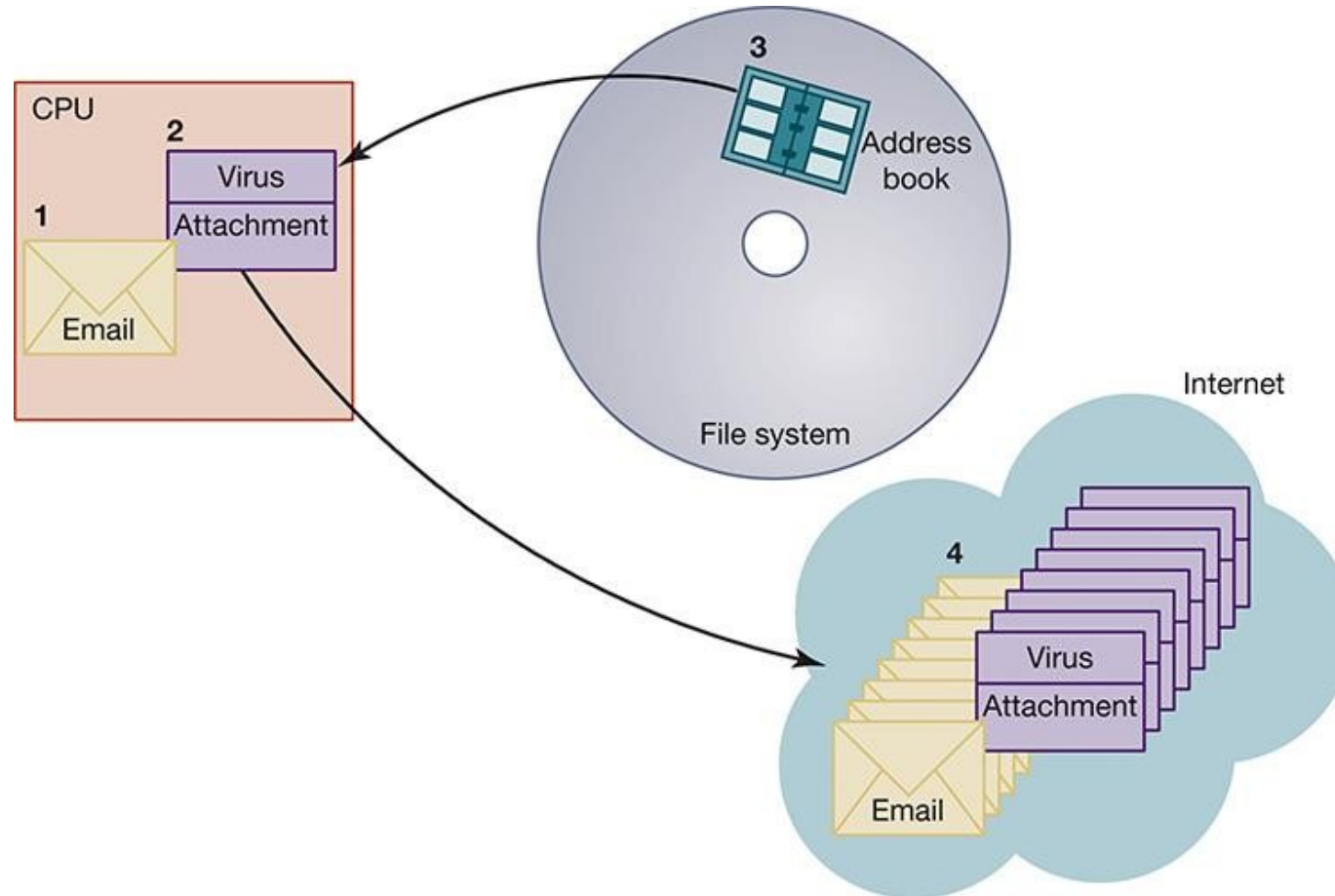


7.3.1 Viruses

- ❖ Virus: Piece of self-replicating code embedded within another program (host)
- ❖ Viruses associated with program files
 - Hard disks, thumb drives
 - Email attachments
- ❖ How viruses spread
 - Thumb drives and Email
 - Files downloaded from Internet



How an email virus spreads



Based on Quinn, M.J. (2020). *Ethics for the information age*. (8th ed.)

7.3.2 The Internet Worm

- ❖ Robert Tappan Morris, Jr.
 - Graduate student at Cornell
 - Released worm onto Internet from MIT computer
- ❖ Effect of worm
 - Spread to significant numbers of Unix computers
 - Infected computers kept crashing or became unresponsive
 - Took a day for fixes to be published
- ❖ Impact on Morris
 - Suspended from Cornell



Based on Quinn, M. J. (2020). *Ethics for the information age*. (8th ed.)

3 years' probation + 400 hours

7.3.3 Sasser

- ❖ Lunched in April 2004,
- ❖ exploited weakness with PCs running the Windows OS
- ❖ Updated software were safe
- ❖ Around 18 million computers worldwide
- ❖ Worm made millions of computers unusable and disrupted operations at Delta Airlines, the European Commission, Australian railroads, and the British coast guard
- ❖ The worm was created by Jaschan, who was 17,
- ❖ After his confession, he was sentenced to worked at a German computer security firm Securepoint



7.3.4 Instant Messaging Worms

- ❖ Robert Tappan Morris, Jr.
 - Graduate student at Cornell
 - Released worm onto Internet from MIT computer
- ❖ Effect of worm
 - Spread to significant numbers of Unix computers
 - Infected computers kept crashing or became unresponsive
 - Took a day for fixes to be published
- ❖ Impact on Morris
 - Suspended from Cornell
 - 3 years' probation + 400 hours community service
 - \$150,000 in legal fees and fines



7.3.5 Conifer

- ❖ Conficker (or Downadup) worm, appeared on Windows computers in November 2008, is notable because of its persistence
- ❖ The worm is able to propagate in several ways
- ❖ Spread to outdated computers with the latest security patches from Microsoft
- ❖ Another version, had two new features that accelerated its spread
 - The ability to invade computers with weak password protection
 - The ability to propagate through USB memory sticks and shared files on local area networks
- ❖ In 2017, two million computers, mostly in India, China, and Brazil, were infected
- ❖ Fortunately, the worm's only purpose seems to be to propagate; it does not steal data or perform other malicious actions. (8th ed.,



7.3.6 Cross-Site Scripting

- ❖ Another way malware may be downloaded without user's knowledge
- ❖ Problem appears on Web sites that allow people to read what others have posted
- ❖ Attacker injects client-side script into a Web site
- ❖ Victim's browser executes script, which may steal cookies, track user's activity, or perform another malicious action



7.3.7 Drive-By Downloads

- ❖ Unintentional downloading of malware caused by visiting a compromised Web site
- ❖ Also happens when Web surfer sees pop-up window asking permission to download software and clicks “Okay”
- ❖ Google Anti-Malware Team says 1.3 percent of queries to Google’s search engine return a malicious URL somewhere on results page



7.3.8 Trojan Horses and Backdoor Trojans

- ❖ Trojan horse: Program with benign capability that masks a sinister purpose
- ❖ Backdoor Trojan: Trojan horse that gives attack access to victim's computer



7.3.9 Ransomware

- ❖ A malware designed to extort money from the victim of the attack
- ❖ It may be installed by different means on a victim's computer
- ❖ Early versions often used a phony message from a law enforcement agency to accuse victims of an illegal activity and to extract “fines” from them
- ❖ The message informed victims that their IP address had been associated with some sort of illegal activity, such as:
 - Downloading child pornography
 - Demanded that they pay a fine using a prepaid

7.3.10 Rootkits

- ❖ Rootkit: A set of programs that provides privileged access to a computer
- ❖ Activated every time computer is booted
- ❖ Uses security privileges to mask its presence



7.3.11 Spyware and Adware

- ❖ Spyware: Program that communicates over an Internet connection without user's knowledge or consent
 - Monitor Web surfing
 - Log keystrokes
 - Take snapshots of computer screen
 - Send reports back to host computer
- ❖ Adware: Type of spyware that displays pop-up advertisements related to user's activity
- ❖ Backdoor Trojans often used to deliver spyware and adware

Based on Quinn, M.J. (2020). *Ethics for the information age*. (8th ed.)



7.3.12 Bots and Botnets

- ❖ Bot: A kind of backdoor Trojan that responds to commands sent by a command-and-control program on another computer
- ❖ First bots supported legitimate activities
 - Internet Relay Chat
 - Multiplayer Internet games
- ❖ Other bots support illegal activities
 - Distributing spam
 - Collecting person information for ID theft
 - Denial-of-service attacks



7.3.12 Bots and Botnets

- ❖ Botnet: Collection of bot-infected computers controlled by the same command-and-control program
- ❖ Some botnets have over a million computers in them
- ❖ Bot herder: Someone who controls a botnet



7.3.13 Security Risks Associated with “Bring Your Own Device”

- ❖ Some companies allow employees to bring their personal smartphones, tablets, and laptops to the workplace and use them for work-related activities
- ❖ The “bring your own device” (BYOD) movement brings numerous benefits
 - Reduce their hardware and software expenditures
 - Use their own devices to access the company network
 - Complete their tasks
- ❖ Thus, introducing security risks, such as:
 - Company data may be compromised as consequences of stolen devices
 - If the employee's device is insecure, it may provide an avenue for a



7.3.13 Security Risks Associated with “BYOD”

- ❖ To address security concerns, companies establish BYOD policies including:
 - What are the security standards for personal devices (password requirements, antimalware packages, etc.)?
 - What applications are employees allowed to run from their personal devices?
 - What is the level of support for personal devices that the company’s IT department will provide?
 - Does the company have the right to erase all data on a personal device that has been stolen?
 - How will company data be removed from the devices of employees who are leaving the company?



7.4 Cyber Crime and Cyber Attacks



7.4.1 Phishing and Spear Phishing

- ❖ Phishing: Large-scale effort to gain sensitive information from gullible computer users
 - At least 67,000 phishing attacks globally in second half of 2010
 - New development: phishing attacks on Chinese e-commerce sites
- ❖ Spear-phishing: Variant of phishing in which email addresses chosen selectively to target particular group of recipients

7.4.2 SQL Injection

- ❖ Method of attacking a database-driven Web application with improper security
- ❖ Attack inserts (injects) SQL query into text string from client to application
- ❖ Application returns sensitive information



7.4.3 Denial-of-Service and Distributed Denial-of-Service Attacks

- ❖ Denial-of-service attack: Intentional action designed to prevent legitimate users from making use of a computer service
- ❖ Aim of a DoS attack is not to steal information but to disrupt a server's ability to respond to its clients
- ❖ Distributed denial-of-service attack: DoS attack launched from many computers, such as a botnet



7.4.4 Internet-of-Things Devices Co-opted for DDoS Attack

- ❖ On 2016, a DDoS attack against domain name service provider **Dyn**
- ❖ Caused Netflix, Twitter, PayPal, Pinterest and others to become unreachable for several hours
- ❖ Dyn provides the important service of translating domain names into numerical IP addresses
- ❖ What made DDoS attacks noteworthy was that it was launched by the Mirai botnet, which consists of malware-infected IoT devices
- ❖ Powerful, executed by as many as 100,000 devices
- ❖ IoT devices are relatively easy for malicious actors to co-opt because many people depends on the devices factory default settings

❖ Based on Quinn, M. J. (2020). *Ethics for the information age*, (8th ed.)
As a result, millions of IoT devices may already be infected by



7.4.5 Cyber Crime

- ❖ Criminal organizations making significant amounts of money from malware
- ❖ Jeanson James Ancheta
- ❖ Pharmamaster
- ❖ Albert Gonzalez
- ❖ Avalanche Gang



7.4.6 Politically Motivated Cyber Attacks

Some nation-states, terrorist organizations, and allied groups are mounting politically motivated cyber attacks on the computer and network infrastructure of their opponents, and some of these efforts have caused major disruptions

- ❖ Estonia (2007)
- ❖ Georgia (2008)
- ❖ Georgia (2009)
- ❖ Exiled Tibetan Government (2009)
- ❖ United States and South Korea (2009)
- ❖ Stuxnet Worm (2009)



Reference

Quinn, M. *Ethics for the information age.*, 8th edition, Chapter 7



Thank
You

